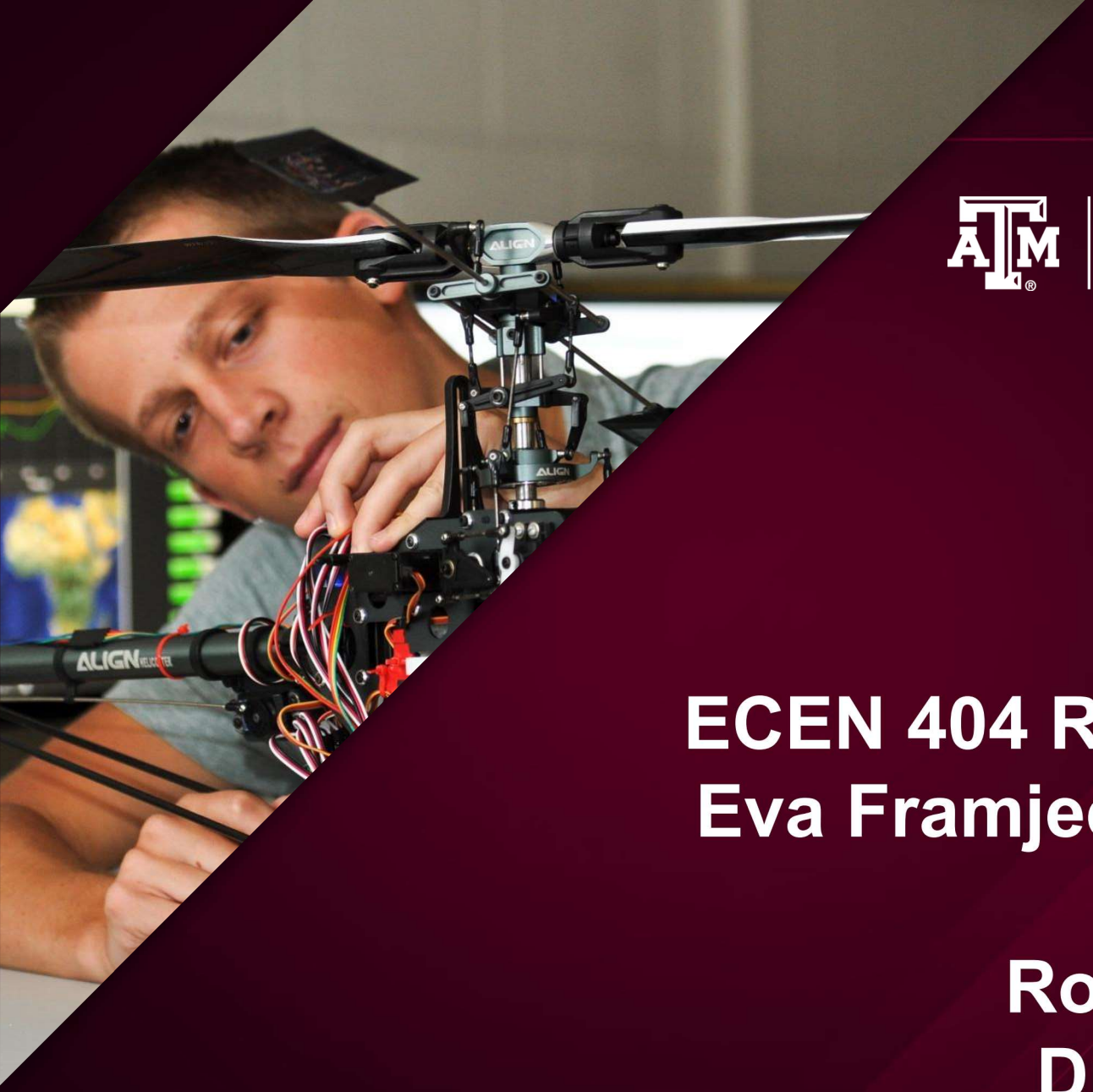


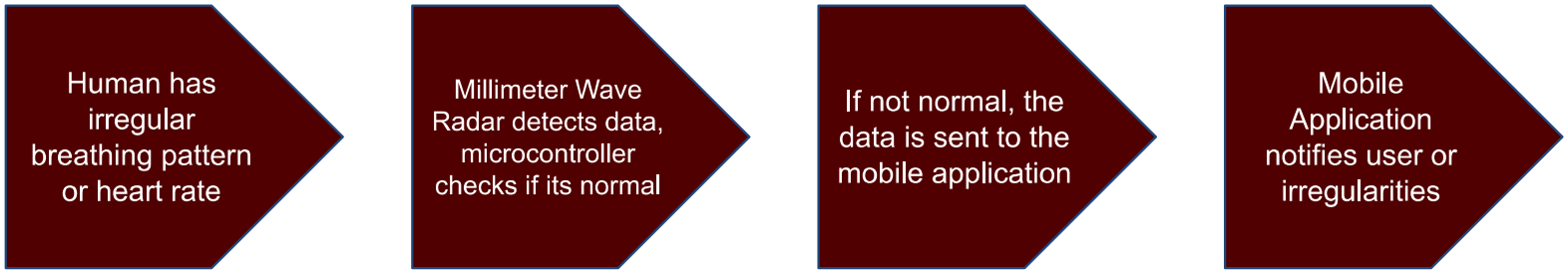
Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Radar Rangers

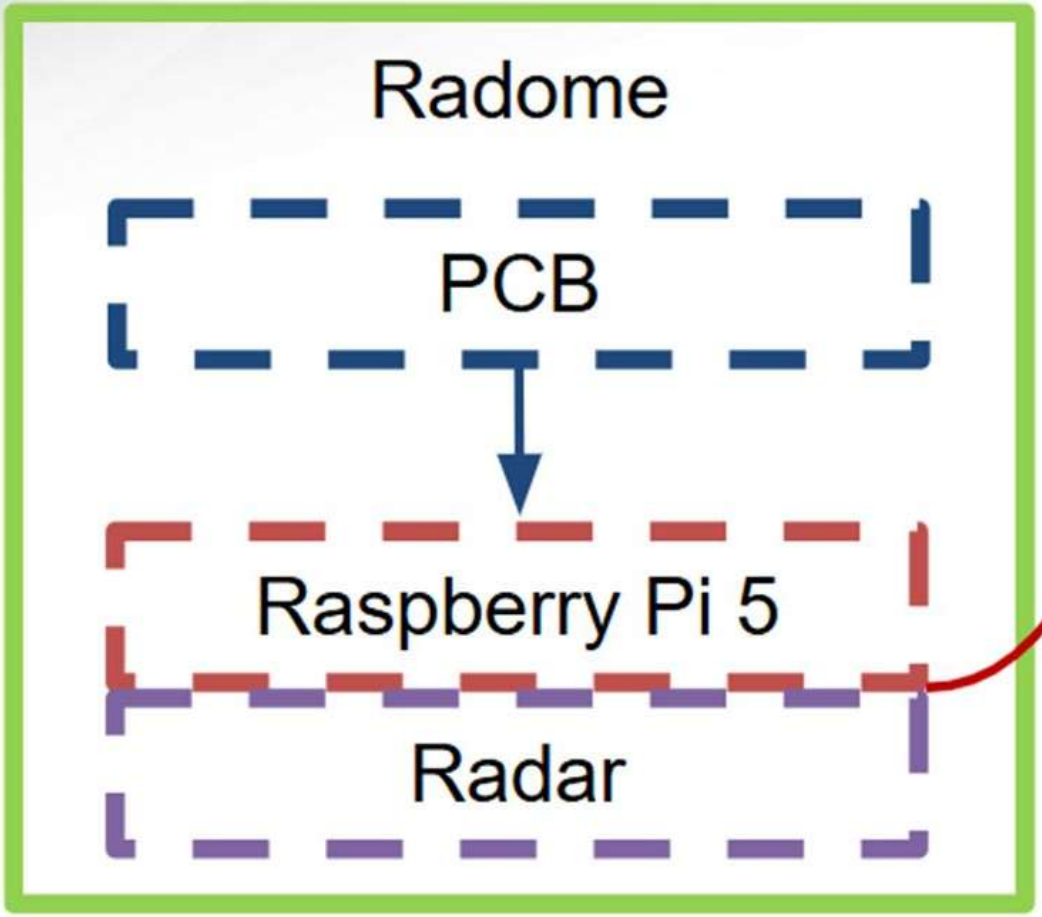
Eva Framjee, Neil Garcia,
Sterling Light
Roman Venegas
Dr. Arya Menon

Problem Overview

- Devices can't detect breathing and heart rate without wearable sensors, causing discomfort.
- Provide a fully contactless method for tracking breathing patterns and heart rate which can be read easily via mobile devices.

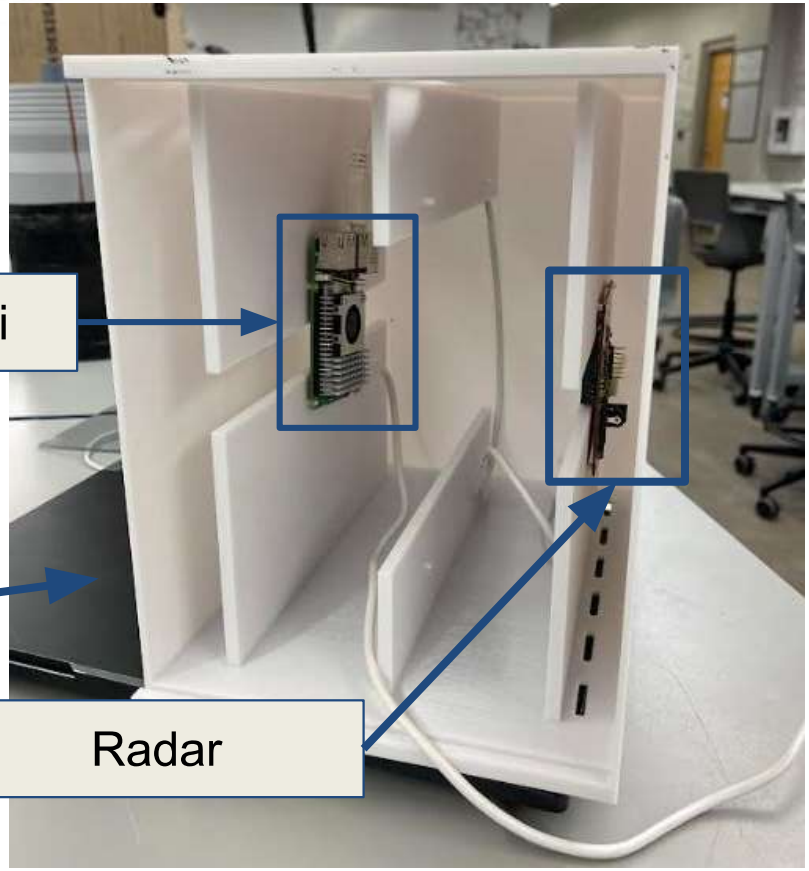


Integrated Project Diagram



Radome & PCB: Sterling
RasPi & Radar: Neil
Mobile Application: Eva

Integrated Project Diagram

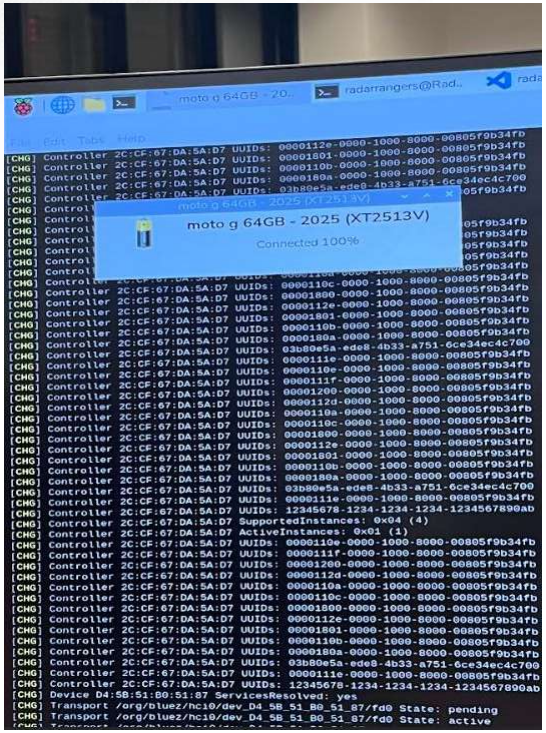


Integrated Project Diagram

https://youtube.com/shorts/qNDttjFyWto?feature=share

Radar and Raspberry Pi Accomplishments

Neil Garcia

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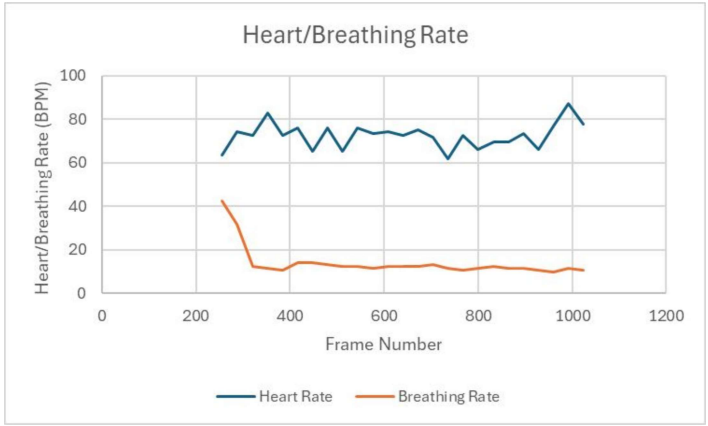


Radar and Raspberry Pi Accomplishments

Neil Garcia

Functional Requirement	Metric of Functionality	Results
Radar accurately detects vitals at reasonable range	• TI IWRL6432Boost radar module is able to measure vitals with +/- 10 bpm of physical measurements. • Maintains accuracy of vitals measurements up to 10 feet away	• Radar module consistently measures vitals within 10 bpm away • PASS
Machine Learning Model is accurate with its vitals decision.	• Random Forest ML model is able to determine vitals state with 0% false negatives and <1% false positives	• Machine learning model has 0% false negatives ~1% false positives. • PASS

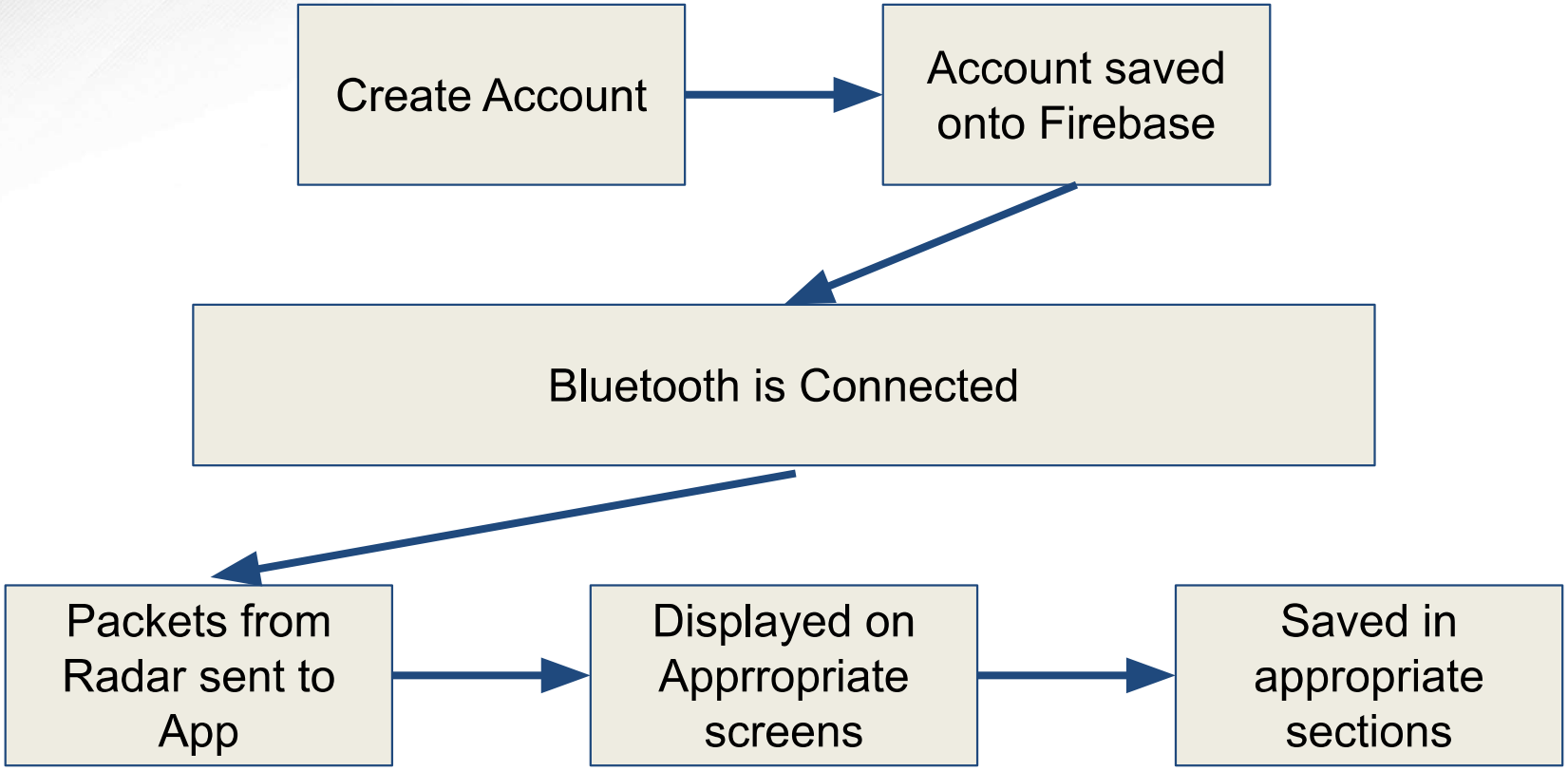
Radar and Raspberry Pi Accomplishments

Neil Garcia

- Challenges
  - Integrating radar module with Raspberry Pi without pre-developed integration software.
  - Determining normal and abnormal vitals with radar vitals irregularities.
  - Sending vitals data to the phone app.
- Solutions
  - Reverse engineered existing software to be compatible with Linux/Raspberry Pi.
  - Used an ML model to filter out irregular frames of data while making vitals decision.
  - Used Bluetooth Low Energy to transfer live data from Pi to phone.

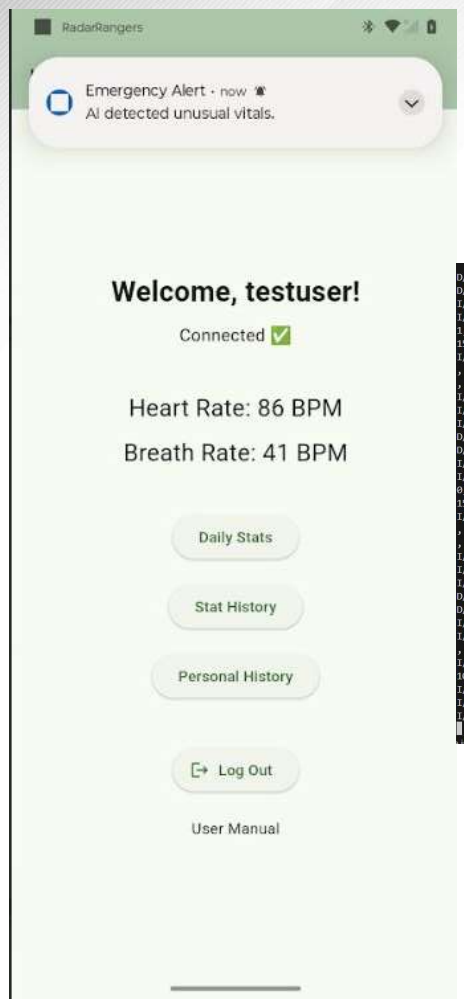
Mobile Application Accomplishments

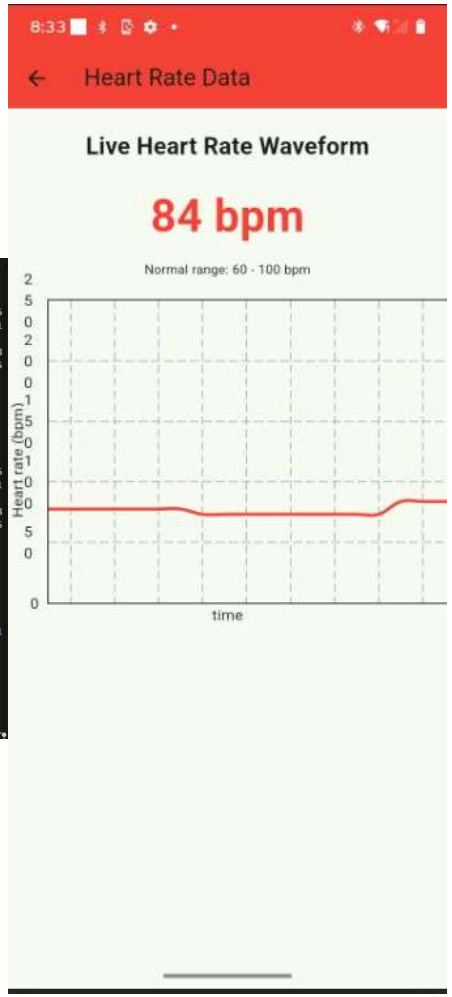
Eva Framjee



Mobile Application Accomplishments

Eva Framjee





Mobile Application Accomplishments

Eva Framjee

- Challenges
  - Making the Phone run the app without being connected to the computer.
  - Connecting the Raspberry Pi to the Phone
  - Storage
- Solutions
  - We set up a connection via bluetooth
  - Set up to save on phone itself

```
PS C:\Eva\TAMU8\ECEN403\Code for login\flutter_application_1> adb devices
List of devices attached
ZY32L7W9SG      device

PS C:\Eva\TAMU8\ECEN403\Code for login\flutter_application_1> adb install -r build\app\outputs\flutter-apk\app-debug.apk

Performing Streamed Install
Success
```

PCB and Radome Accomplishments

Sterling Light

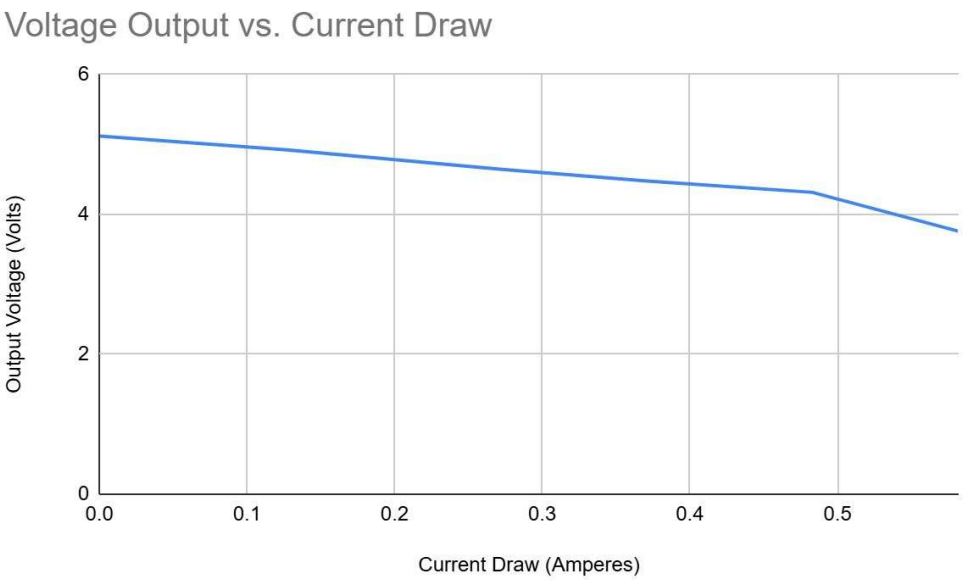
Functional Requirement	Metric of Functionality	Results
Radome properly holds boards in place	• Radome can hold IWRL6432BOOST, PCB, and Pi 5 in place for >30 minutes without intervention • Boards remain in place when radome is subjected to physical adjustment	• Radome held boards in place for >1 hour • Boards remained mounted when radome was held facing downward and shaken and struck • PASS
Radome is electromagnetically transparent for radar	• With radome integrated, radar heart rate reading stays within 10 bpm of oximeter reading	• With radome integrated, heart rate of radar remained within 10 bpm of oximeter reading • PASS
PCB reliably powers Pi 5 and IWRL6432BOOST board	• PCB provides enough power for Pi 5 and IWRL6432BOOST board to start up • Radar operates normally without incident for >30 minutes	Untested

Radome Status

Sterling Light

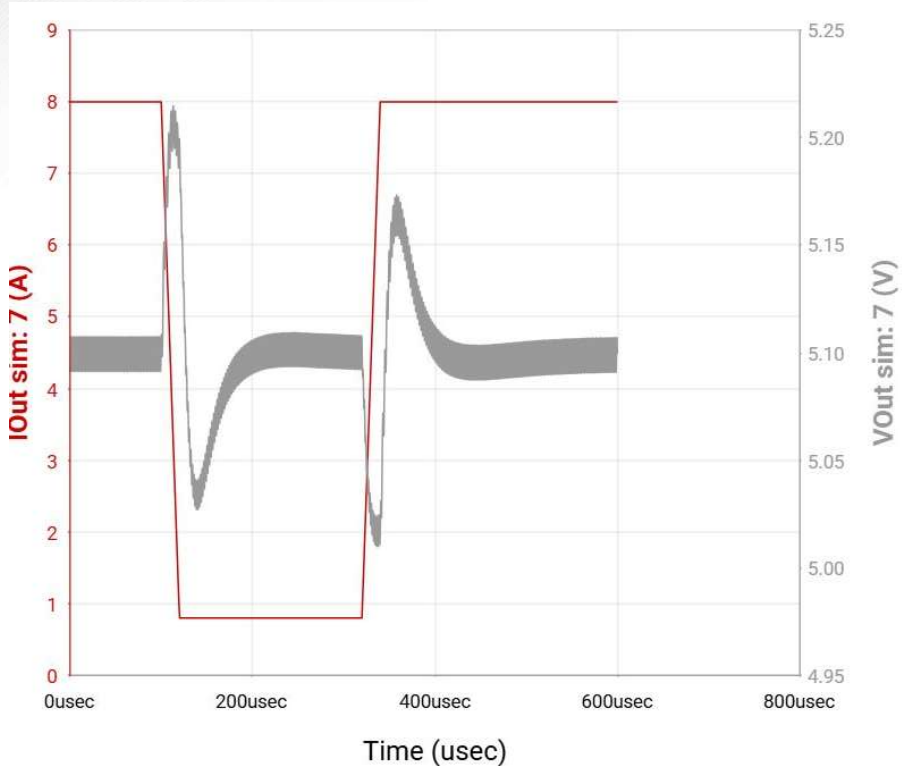
PCB Status

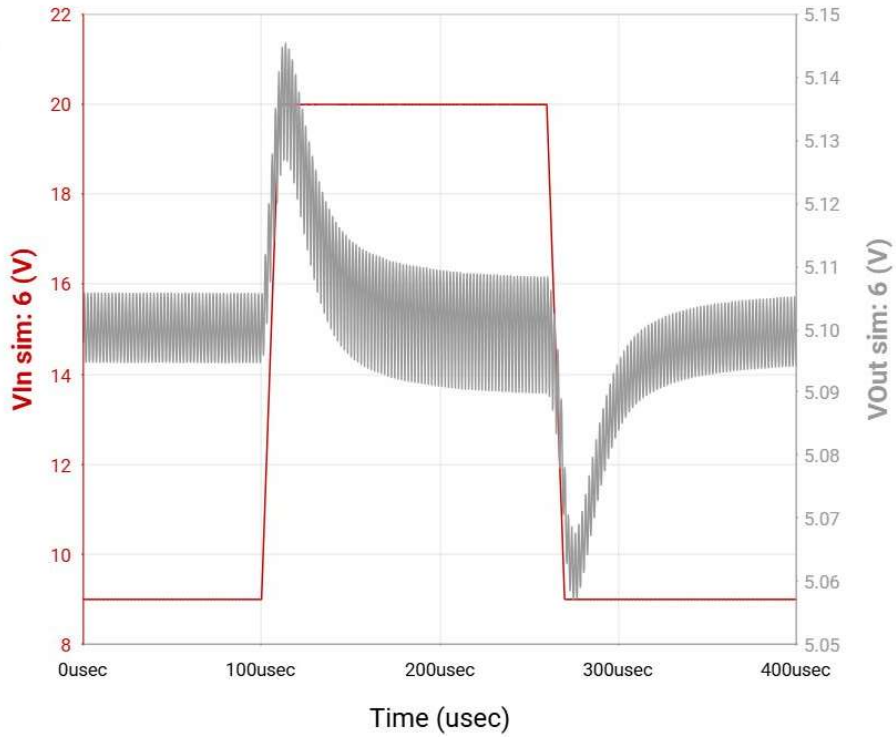
Sterling Light



PCB Status

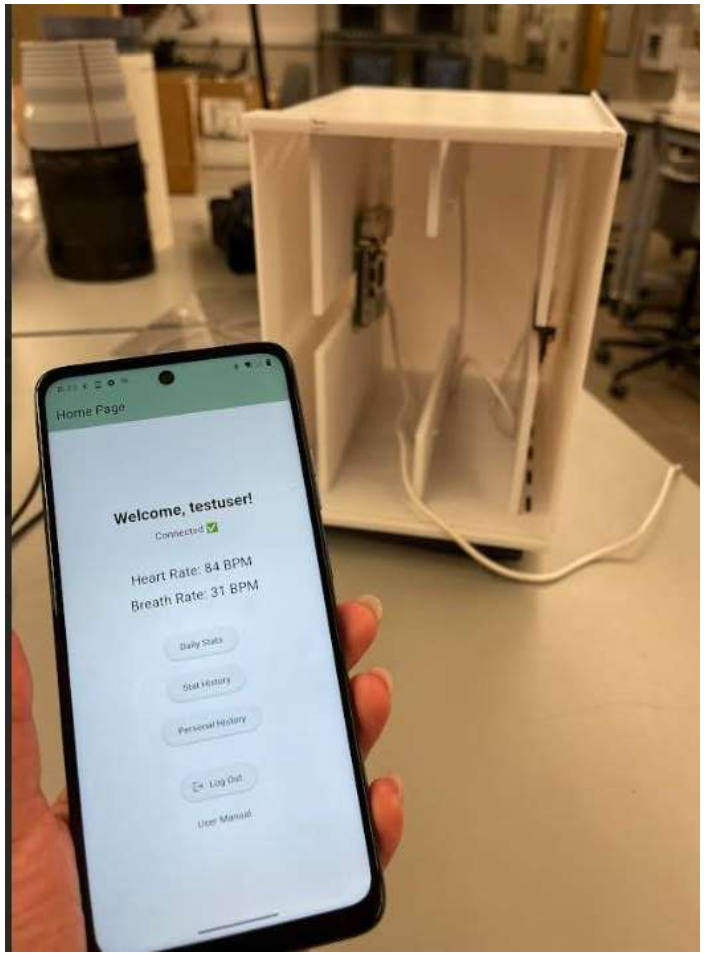
Sterling Light





Integrated System Results

- Test Cases
  - Sleeping/Resting
  - Speaking
  - Error when multiple people are in room
  - Through the Radome
  - Without Radome
- Waiting for PCB
  - Test Cases
    - Radar system is on but without a target
    - Radar system is on and actively taking measurements from a target



Conclusions

- Issues encountered:
  - Original PCB insufficient as power supply
  - Radome design had to be altered to properly contain components
  - Current PCB encountering issues during testing
  - Headless connection between the Ras Pi and the phone
  -
- Current Status:
  - Running test cases with the device
  - Rectifying issues with current PCB

Thank You!